

Untracked Nightmares: The Threats Hiding Behind Commodity Infrastructure

Executive Summary

A recent Unit 42 investigation into seemingly low-priority enterprise infections demonstrates how the most effective camouflage in cybercrime is not necessarily in the use of sophisticated techniques, but in how unremarkable the threat appears. The activities that we investigated would typically not require escalation or further inquiry. But upon closer inspection, we discovered a massive cybercrime campaign largely targeting young gamers. Tracked as CL-CRI-1171, in accordance with [Unit 42's attribution framework](#), the group behind this cluster has operated under the radar for at least two years, distributing an indeterminate number of payloads.

The group behind CL-CRI-1171 provides an infection service for other threat actors who want to spread their malware indiscriminately. This pay-per-install (PPI) marketplace drove hundreds of infections through YouTube channels and a parallel search engine optimization (SEO)-poisoning funnel, all using the same custom loader.

We observed at least eleven YouTube channels that had hundreds of thousands of followers. We notified YouTube of these channels, which it promptly terminated.

These channels were actively interacting with viewers to promote gaming content laced with links to download malware. Content in the channels included advice on improving frame rates, fixing game crashes and adjusting settings on game platforms. Although the videos provided real content for gamers, they also served as the delivery vehicle for infection, prompting viewers to download malicious tools.

The SEO funnel targeted a more professional audience, promoting trojanized software that resulted in malware deployment on corporate endpoints, including critical infrastructure and even government entities. We identified three independent payloads delivered by the same loader between July 2025 and

April 2026: two never publicly reported, Docro Hijacker and ARKTunnel, and a new variant of a previously unnamed backdoor, which we dubbed Insomnia remote access Trojan (RAT).

These infections represent only a small sample of a much larger deployment campaign. We have identified more than 10,000 distinct loader samples, each capable of delivering unique payload combinations.

We provide an overview of the cybercrime cluster and its loader infrastructure, and a technical analysis of three recently delivered malware strains.

Palo Alto Networks customers are better protected from the threats discussed in this article through the following:

- [Advanced WildFire](#)
- [Advanced URL Filtering](#) and [Advanced DNS Security](#)
- [Cortex XDR](#) and [XSIAM](#)

If you think you might have been compromised or have an urgent matter, contact the [Unit 42 Incident Response team](#).

Related Unit 42 Topics	SEO Poisoning , Browser Hijacking , RATs
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Overview of CL-CRI-1171 Activity

Our discovery of two separate infections delivering three entirely distinct malware families revealed one common denominator: a shared loader. By tracing this infrastructure, we mapped the broader activity of CL-CRI-1171, ultimately tying the cluster to a PPI marketplace responsible for delivering countless payloads over the last two years.

This operation uses at least two funneling mechanisms to route traffic to the malware landing pages: a network of YouTube channels and [SEO poisoning](#). The shared infrastructure between the YouTube

and SEO funnels, consistent use of the same loader and a rotational domain pattern observed over an eight-month window all pointed to a single sustained operation, which we track as CL-CRI-1171.

The PPI Ecosystem: An Infection Marketplace

PPI networks operate as underground marketplaces. An operator compromises machines and auctions access to multiple buyers, each buyer deploying their own independent payloads through the same dropper. The result is a single infection that conceals multiple payloads from unrelated threat actors on the same endpoint, each with its own C2 infrastructure and objectives. Because the loader is designed to be disposable and generic, it rarely attracts the scrutiny needed to uncover its subsequent payloads.

Tracking CL-CRI-1171: The PPI Operation Behind the Payloads

The investigation began with two identical and seemingly routine infections at separate organizations. Both events involved trojanized software downloads — one a Bluetooth driver, the other [WinDirStat](#) — and both affected machines executed an identical post-exploitation chain. The loader was unnamed, untracked and generic enough to be dismissed as commodity adware.

But our discovery of a shared payload chain raised a question. How were two unrelated organizations infected by the same unnamed loader delivering the same set of payloads, just one week apart?

Pivoting on the loader's C2 infrastructure revealed a sprawling network of rotational domains — over 200 unique hostnames following a distinctive two-word compound naming pattern (including `bubbleslip`, `churchpail`, `dinosaursjam`), rotating across `.xyz`, `.cfd`, `.space` and `.info` top-level domains (TLDs).

Unpacking the Delivery Infrastructure

The payloads delivered through CL-CRI-1171's infrastructure are not fixed, enabling the simultaneous delivery of unrelated malware families. In April 2026, two incidents initiated this investigation. They shared a single loader that delivered three separate malware families: Insomnia RAT, ARKTunnel and Docro Hijacker. A subsequent infection, observed in June 2026, delivered two entirely different

payloads: GCleaner and Socks5Systemz.

Figure 1 illustrates the structure of the operational architecture and the observed rotational malware bundles.

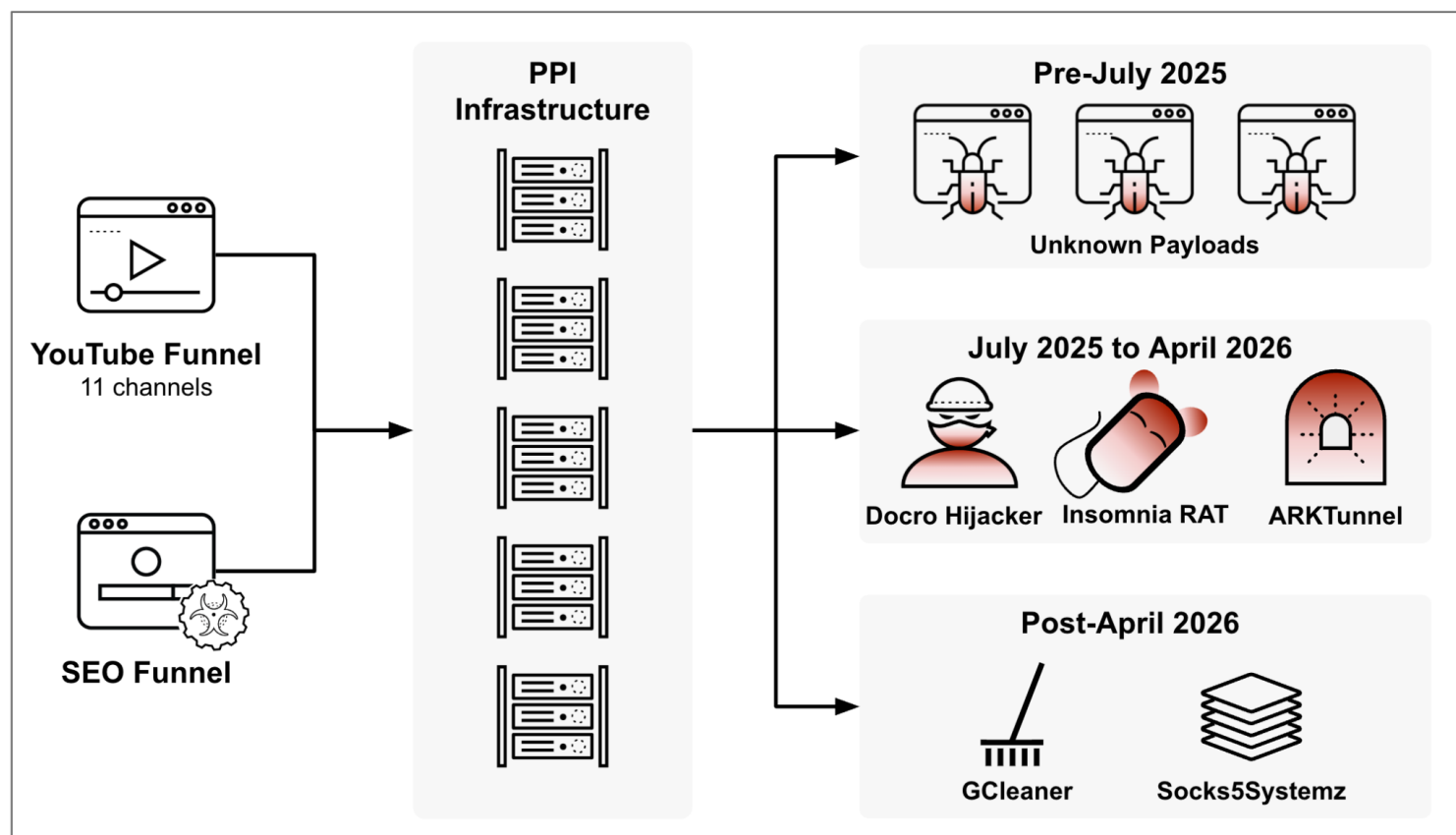


Figure 1. Illustration of CL-CRI-1171 infrastructure.

Our in-depth analysis explores three operations that were spotted in two intrusion sets during April 2026. By examining the campaign's trackers, we found that these operation payloads were in rotation from July 2025 to April 2026. This discovery provides a valuable snapshot of the group's deployment capabilities and the variety of payloads they have been entrusted to deliver.

Although the observed rotation appears to be highly structured, the discovery of over 10,000 unique OfferLoader samples points to a much larger distribution pipeline. It is very likely that the loader has delivered numerous other malware families.

The SEO poisoning path was the first delivery channel we identified. Unsuspecting users searching for legitimate software landed on file-hosting lure pages that displayed a fake virus-scan animation before delivering the trojanized archive via a chain of redirectors into the PPI gate.

A reconstructed example download page from the SEO poisoning funnel is shown in Figure 2.

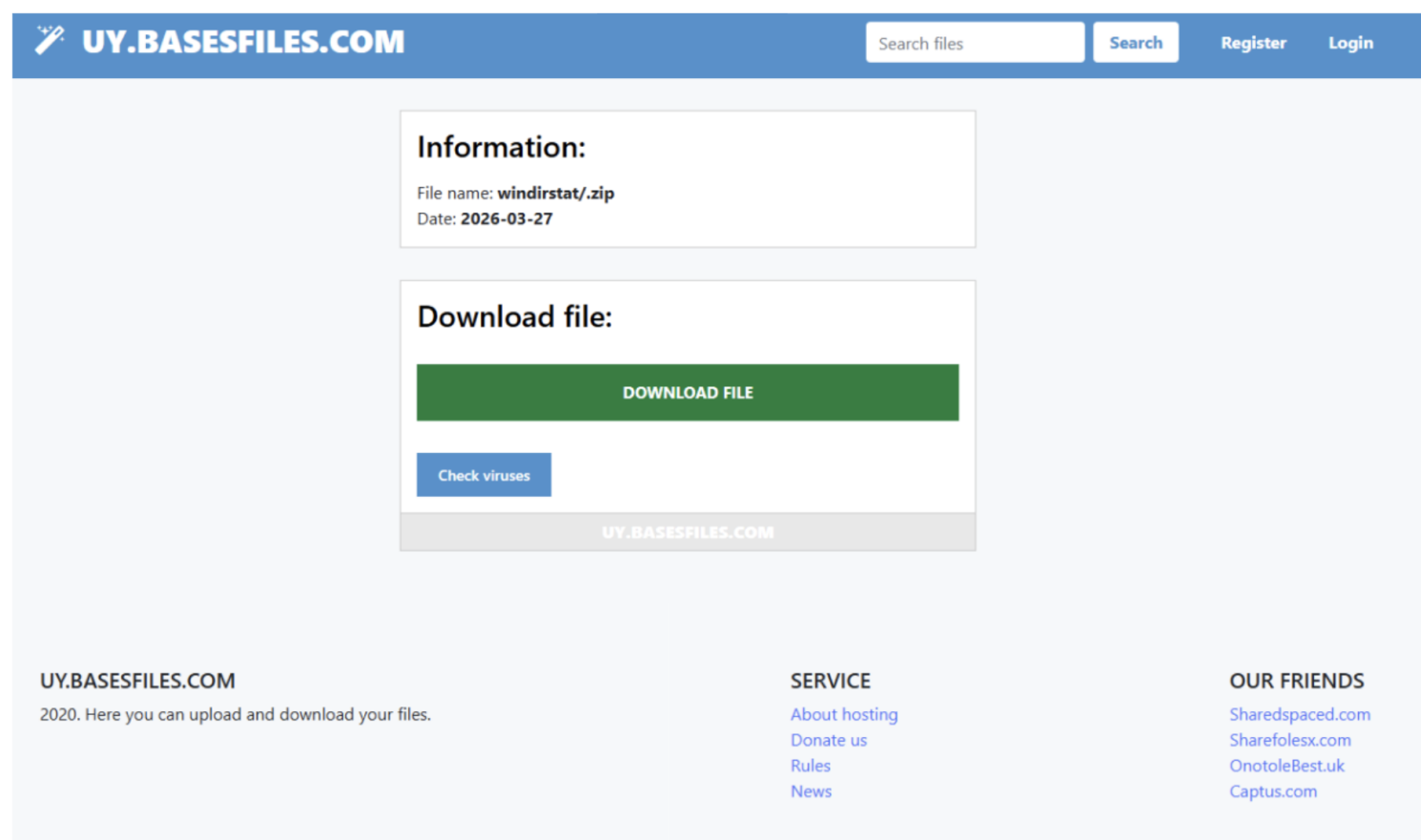


Figure 2. Reconstruction of the trojanized WinDirStat download page.

Analysis of the gate's tracker URLs revealed a critical detail. Each URL contained a `click_id` parameter: a Base64-encoded fingerprint containing the victim's operating system (OS), browser, the referring domain, the exact search keyword that led them to the lure and their public IP address, as Table 1 shows.

Operating System	Browser	Referring SEO Domain	Search Keyword	Victim IP
Windows_10	Chrome	atthelake[.]info	hwidspoofer	5.xxx.xx.xxx
Windows_10	Chrome	atthelake[.]info	combatwarriorsGit	2.xx.xxx.xx

Table 1. Example of a deobfuscated click_id.

The gate uses this fingerprint to decide who receives the payload: a valid, fresh click_id forwards the visitor to the malware/loader download. This is an evasion tactic used to ensure that only real targets are infected, and to protect the actor’s infrastructure: scanners, crawlers or analysts receive a

decoy clone of the legitimate WinRAR download page or broken links. This is why the campaign has almost no public footprint despite being highly active: automated URL scanners rarely pass the gate.

Decoding hundreds of these fingerprints, with the assistance of AI, exposed the full names of YouTube video titles and their associated channels. Alongside search keywords for cracked software and game cheats, the `q=` field – which tracks which search query led to the infection – contained strings like "CS2 Potato Graphics Settings - Maximum FPS Boost for Low-End PC (2026 Guide) - Velvox."

The YouTube Funnel

Searching for titles derived from the fingerprints led us to uncover a chain of gaming optimization YouTube channels. Their content included tips on boosting frames per second (FPS), fixing game crashes and tweaking settings on popular game platforms. While the videos provided authentic, engaging content for young gamers, they ultimately served as an infection delivery vehicle, prompting viewers to download a malicious “tool” or “optimization pack” via links in the description, as Figure 3 shows.

Your CPU Is Holding Back Your GPU — Fix CPU Bottleneck & Low GPU Usage (2026)



ADEX

18.7K subscribers

Subscribe

168 views Jun 22, 2026 #CPUBottleneck #LowGPUUsage #GPUUsageFix

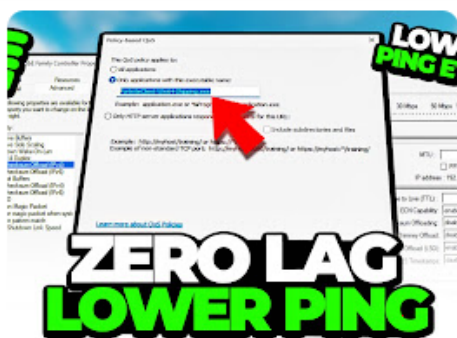
Your CPU Is Holding Back Your GPU — Fix CPU Bottleneck & Low GPU Usage (2026)

✓ Boost Pack Link: <https://bit.ly/3zLjHXt>

✓ Get The GearUp Booster FREE: <https://aff.gearupglobal.com/product/...>

Figure 3. A YouTube video from the ADEX channel directing viewers to a download link.

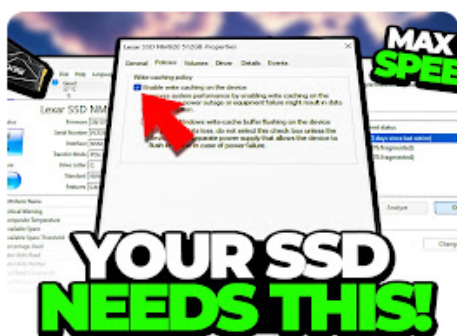
Those links pass through intermediary sites, such as Blogspot, which contain social-engineering instructions that lead the victim to the same PPI gate infrastructure serving the SEO path. An example blog page is shown in Figure 4.



Fix High Ping & Packet Loss NOW! 🚀 Zero Lag, Lower Ping & Faster Downloads

ALPHA NOTCH FPS BOOST JULY 04, 2026

To get started, click the Download button below. You'll be redirected to a quick unlock step where you can subscribe to gain...



Your SSD Is Holding Back Your PC — Here's Why & How to Fix It! (2026)

ALPHA NOTCH FPS BOOST JULY 02, 2026

To get started, click the Download button below. You'll be redirected to a quick unlock step where you can subscribe to gain...

Figure 4. The download link leads to a Blogspot page.

We identified 11 channels connected to CL-CRI-1171, which collectively had hundreds of thousands of subscribers and millions of views.

Technical Analysis

Our investigation revealed three malware strains delivered by the group behind CL-CRI-1171 between June 2025 and April 2026. The following sections explain the technical aspects of the loader used by this cybercrime group and the deployed malware payloads:

- **OfferLoader** – The delivery mechanism behind all recorded intrusions, an Inno Setup trojanized installer that sets up the other payloads.
- **Operation A: Insomnia RAT** – A dual-payload, cross-platform backdoor that brings its own environment to ensure survival. Targeting both Windows and macOS using Node.js, paired with a twin Python agent.

- **Operation B: ARKTunnel** – A previously unreported WebSocket tunneling RAT unpacks itself from a bitmap image using steganography. We found 50 samples spanning over a year of development, operating across four fictitious corporate-identity rotations.
- **Operation C: Docro Hijacker** – A Chrome backdoor that revives a [browser-hijacking technique](#) first seen in 2015, re-engineered to bypass modern integrity protections. This campaign represents the first observation of this modern variant in the wild.

Initial Access Vector in Intrusions

We discovered two intrusion sets that began in the same way: a user searched for a legitimate utility, clicked a top search result that led to a malicious domain, and downloaded what appeared to be a legitimate application. Both infections delivered the same three payloads.

- **First intrusion set:** Downloaded Bluetooth Driver for Windows 10.exe from a file-sharing archive. The installer was a trojanized Inno Setup package carrying the PPI loader with affiliate ID CID=2855.
- **Second intrusion set:** Browsed to noiseship[.]cfd, a domain registered just 39 days earlier, and downloaded a trojanized windirstat.exe installer. This package carried the PPI loader with affiliate ID CID=3075.

The OfferLoader Execution Chain

The operator's code and C2 communication designate each payload slot as an "offer," tracking variables as `offer_execution`, `offer_execution_fail` and `offer_exists`. Based on the naming convention, we track this loader as OfferLoader. OfferLoader uses chained Inno Setup (a legitimate installation packager) packages to deliver multiple payloads. Figure 5 shows the infection chain.

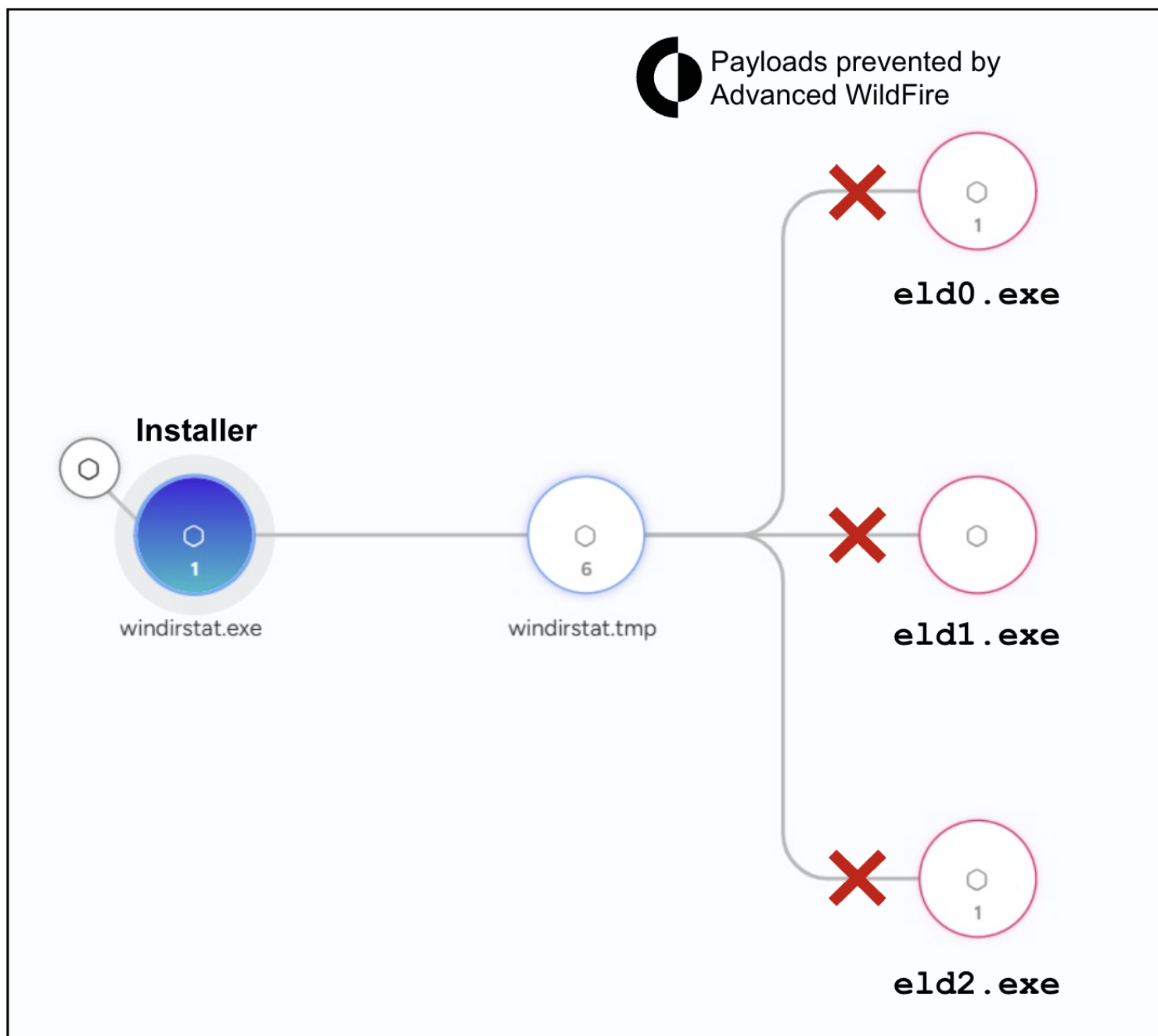


Figure 5. Cortex XSIAM view of the infection chain. In this case, Advanced WildFire blocked the payloads.

OfferLoader is delivered in a ZIP file, with the source download site providing the social engineering instructions required to guide users through downloading and executing an extraction tool. Based on our analysis of the collected samples, we observed the loader being delivered alongside a legitimate version of WinRAR, renamed to **.store**. The installer contains no embedded application files; all malicious logic is contained in the compiled Pascal [Code] section, which triggers when the

installation page is displayed.

In two separate intrusion sets we discovered that the ZIP file contained OfferLoader masquerading as a `windirstat.exe` installer that initiates the compromise by unpacking `windirstat.tmp`. This temporary file transmits an initial tracking beacon to `voyagemist[.]space`. This is another gating mechanism: depending on the structure of the beacon, one of two text files will be retrieved. They either contain “no” to signal that the loader should not unpack further stages, or “ok” to signal all offers will be deployed. Following this check-in, the process spawns three child processes:

`eld0.exe`, `eld1.exe` and `eld2.exe`. Each child process corresponds to a different malware campaign. The loader passes specific affiliate-tracking parameters to each process via a command line.

Figure 6 illustrates the entire OfferLoader infection chain at the time of the intrusion.

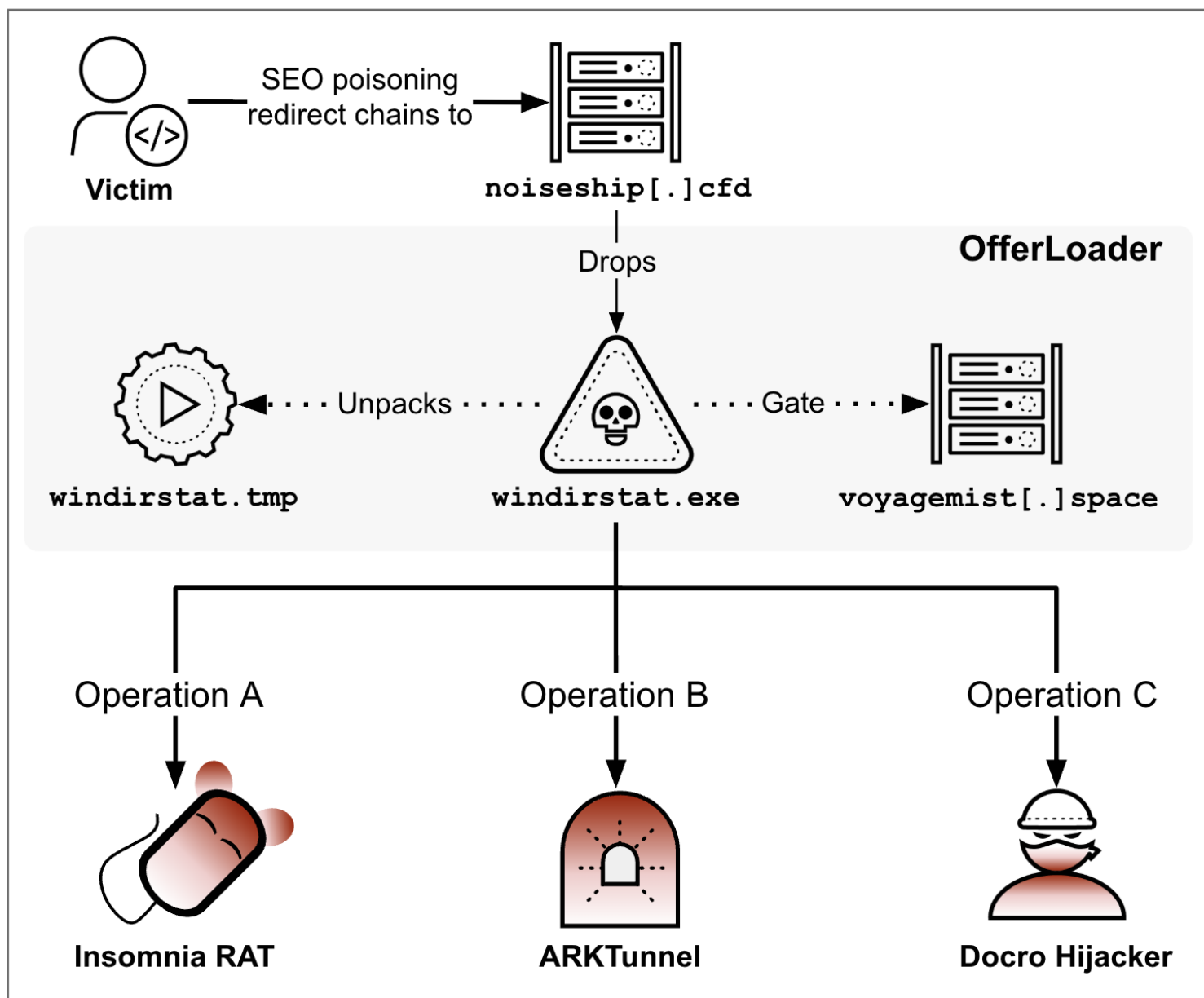


Figure 6. Example of an OfferLoader infection chain.

OfferLoader's role ends once the three offers are spawned. From this point forward, each child process operates as an independent malware campaign with its own infrastructure, C2 protocol and objectives. The following sections analyze what each offer delivers.

Operation A: Insomnia RAT – A Cross-Platform Backdoor

Insomnia RAT simultaneously distributes two payloads:

- An upgraded variant of a Node.js backdoor (reported by [Walmart Global Tech](#) in 2025)
- A complementary Python backdoor

We have dubbed these twin payloads Insomnia RAT, due to the user-agent string used for C2 communications: `insomnia/2023.4.0 Windows`.

Figure 7 shows the full infection chain that delivers Insomnia RAT's twin payloads.

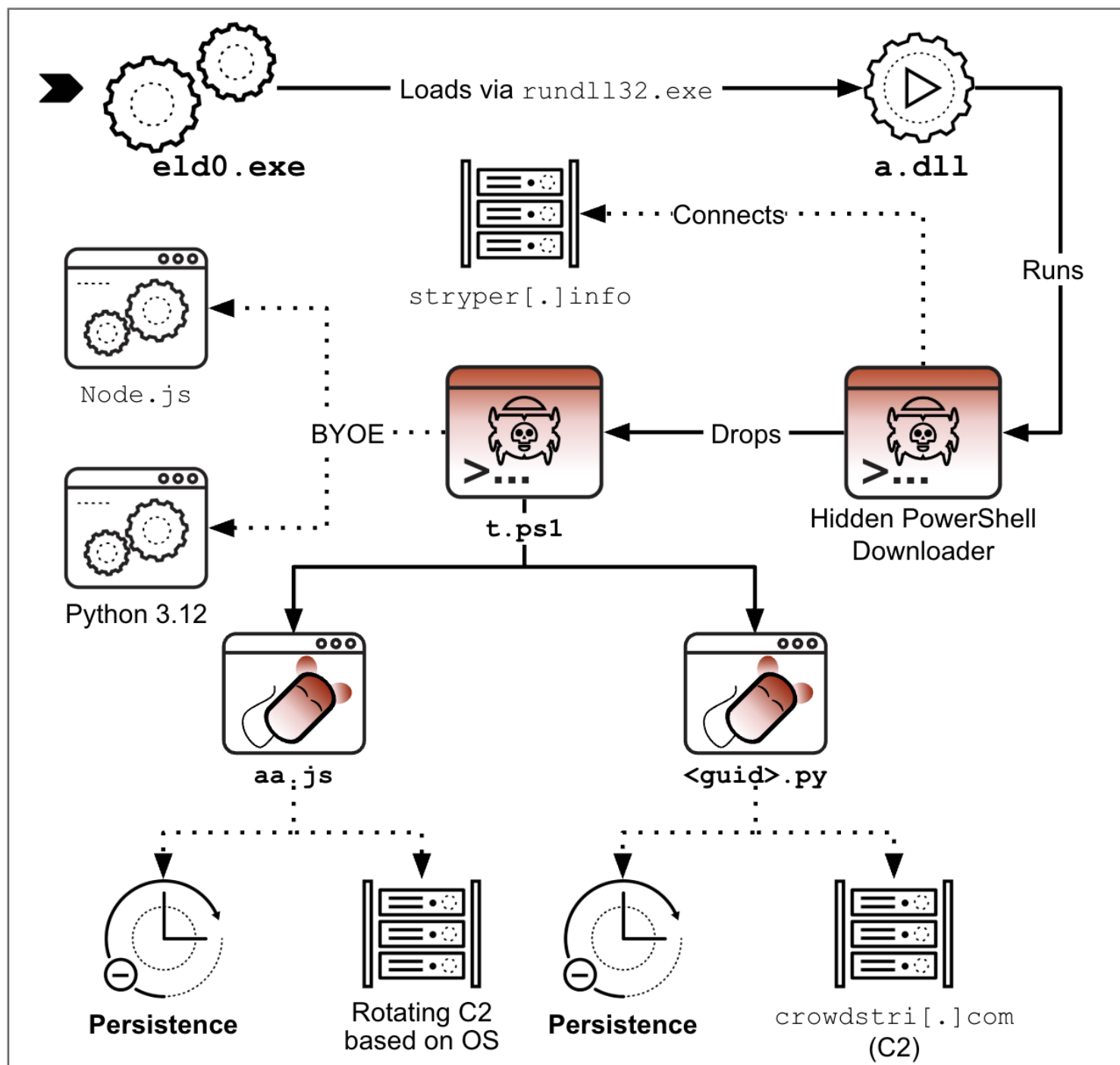


Figure 7. The new Insomnia RAT variant infection chain.

Eld0.exe drops the **a.dll** payload and spawns a hidden PowerShell process to download and execute **t.ps1**. This double-stage installer script disables Windows Defender protections, adds the entire **C:** drive as an exclusion, suppresses security notifications and deploys the two Insomnia RAT backdoors. **t.ps1** also downloads and sets up the environment needed to execute both backdoors by

installing Python and Node.js on the victim's machine. Node.js is hidden from the system's Add/Remove Programs list by setting `SystemComponent=1` in the registry.

The first part of Insomnia RAT is a Node.js agent downloaded from `stryper[.]info/aa.js`. While the prior variant targeted Windows, Linux and FreeBSD with a single payload, this iteration targets Windows and macOS using platform-specific C2 server lists. It also deploys a companion Python agent as a redundant fallback, ensuring persistent access if one runtime environment is detected or removed.

The backdoor collects the victim's MachineGuid, universally unique identifier (UUID), hostname and operating system details. It then contacts its C2 servers via an HTTPS POST request to `/d` using a User-Agent string, `insomnia/2023.4.0 Windows` to disguise itself. The C2 server responds with a JSON array containing commands. These specify a payload type (`node`, `cmd`, `ps1`, `sh`, or `ow` for a self-update) and a download URL. Results are reported back to the `/e` endpoint.

To establish persistence, the installer registers a scheduled task named `Maps Performance Task` under `\Microsoft\Windows\Maps\`. This mimics a legitimate Windows task, executing the backdoor hourly and at system startup under the SYSTEM account.

The same `t.ps1` script installs a second, redundant agent. This Python script is downloaded from `aa.amazingshield[.]xyz`. The installer downloads a legitimate Python distribution and registers a second scheduled task, `OOBETaskScheduler`, under `\Microsoft\Windows\Servicing\`.

The Python agent is simpler than its Node.js sibling, but follows the same C2 protocol pattern: POST requests to `/d` for tasks and POST requests to `/e` for error reporting. The agent collects the MachineGuid, OS product name, hostname and processor architecture.

The Python agent used `crowdstri[.]com` as its C2 domain. This appears to be a deliberate typosquat of `crowdstrike[.]com`, designed to blend into logs and evade quick security reviews.

Operation B: ARKTunnel – A WebSocket RAT Hidden in a Bitmap

The `eld1.exe` chain terminates in a previously undocumented tunnel payload. The chain uses [least-significant-bit \(LSB\) steganography](#) to deploy the final payload. Figure 8 shows the full infection chain of Operation B.

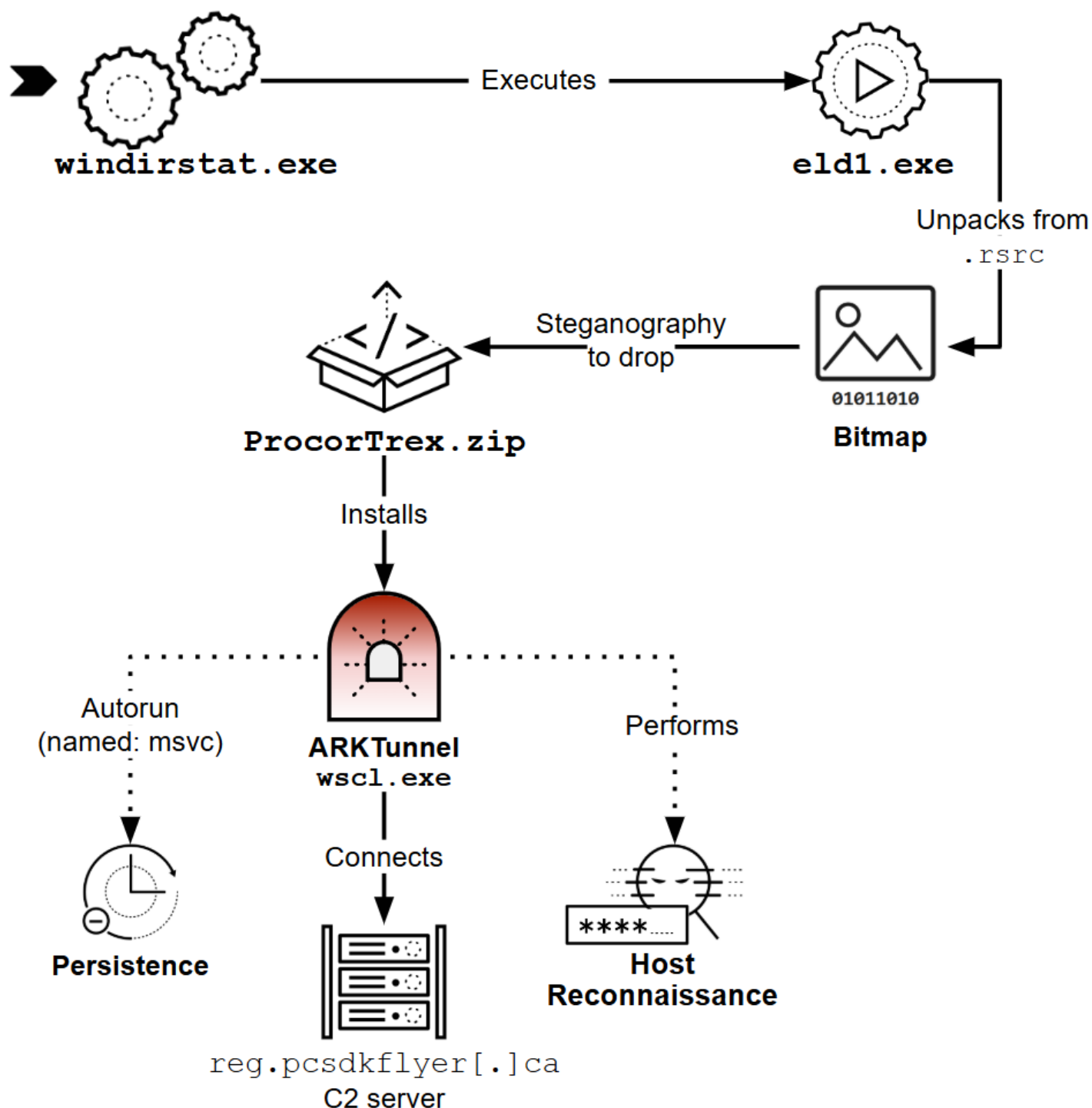


Figure 8. The ARKTunnel infection chain.

Rather than dropping an executable directly to disk, `eld1.exe` extracts a ZIP archive from a BMP image resource using LSB steganography. The result is the payload archive, `ProcorTrex.zip`, which contains `wsc1.exe`, a previously unreported WebSocket-based tunneling RAT. We named this RAT ARKTunnel, based on the attacker's fabricated company name rotation of EarthKark and TamarkLark.

ARKTunnel installs itself as a Windows service named `wsc1-13` or `msvcsvrvc` with a delayed autostart configuration. The RAT supports TCP and UDP tunneling, as well as file execution.

The C2 server address, `reg.pcsdkflyer[.]ca`, is decoded from a 39-byte configuration blob using Base64-decoding followed by an XOR decryption routine. The portable executable (PE) metadata of `wsc1.exe` contains a fabricated company name, **TamarkLark Corporation**, and a fictitious product name, **TamarkLark Manager**, which led us to suspect that other ARKTunnel samples might use additional fabricated company names.

The investigation revealed at least 50 samples deployed over the course of a year. The developer rotated through at least two fake company identities while maintaining an identical icon, binary structure and deployment pattern:

- **EarthLink** in May 2025: EarthLink is the name of a legitimate internet provider, however the attackers used this name coincidentally in the file version information. The attackers did not use or impersonate EarthLink resources or identity.
- **EarthChain** from May 2025–April 2026: EarthChain is also real company name that the attackers used coincidentally. Likewise, also they did not use or impersonate the company's resources or identity.
- **EarthKark**: A fake identity, used from February 2026–June 2026
- **TamarkLark**: A fake identity, used from March 2026–June 2026

All of the samples share the same `wsc1.exe` filename pattern and GUID-based temp directory extraction pattern, indicating that they are variants of the same family.

Despite 50 samples spanning a full year of development and four identity rotations, ARKTunnel has attracted no public reporting or dedicated tracking, with each sample individually flagged as a generic Trojan rather than recognized as a tunneling implant.

Operation C: Docro Hijacker – Reviving Old Techniques

The `e1d2.exe` payload installs a Chrome browser hijacker that we have named Docro Hijacker. The hijacker revives a [browser-hijacking technique](#) that has resurfaced periodically since 2015, now re-engineered to bypass updates to Chrome's integrity protections. While this iteration closely mirrors a proof-of-concept detailed by [Synacktiv](#) in 2025, this campaign marks the updated technique's first documented instance of in-the-wild deployment.

Figure 9 shows the Docro Hijacker installation chain.

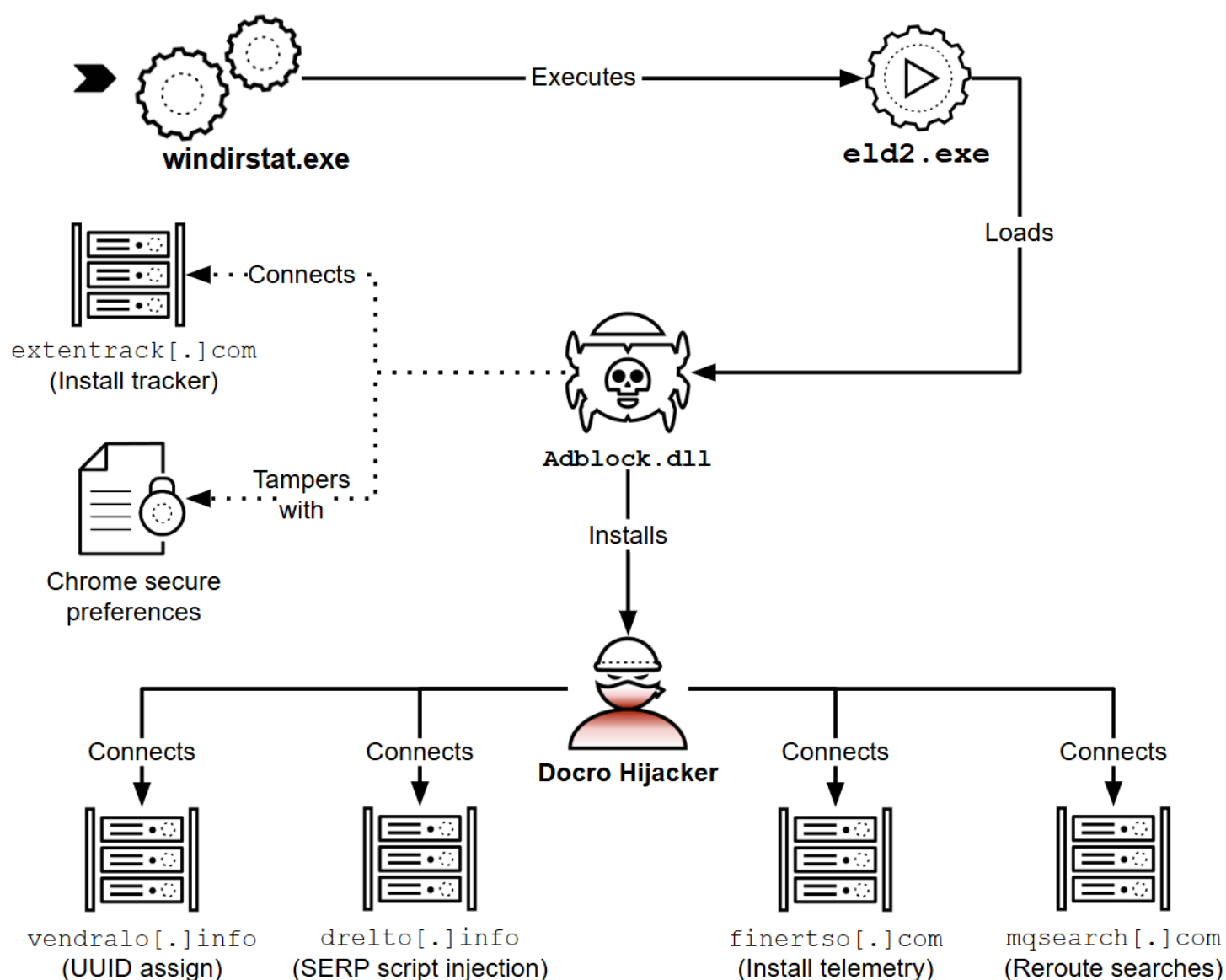


Figure 9. The Docro Hijacker infection chain.

`eld2.exe` is an Inno Setup package, and much like its loader, it extracts and runs `eld2.tmp` which contacts the affiliate's `extentrack[.]com` install tracker.

`eld2.tmp` drops and loads `Adblock.dll`, which bypasses Chrome's Secure Preferences HMAC-SHA256 integrity check. The DLL extracts Chrome's HMAC key from `resources.pak`, computes valid HMAC signatures for the modified preference values and writes them directly to the `Secure Preferences` file. This file functions as an anti-tamper mechanism for browsers by storing a validated copy of the user's settings.

This manipulation allows the malware to execute two primary actions:

- **Search hijacking:** Changes the default search provider to `mqsearch[.]com`, a domain that masquerades as a search engine
- **Extension installation:** Installs the `docro` extension, a Chrome Manifest V3 extension located at `C:\ProgramData\DocsHelper\docro\`

The `docro` extension uses Chrome's `declarativeNetRequest` API to dynamically rewrite network requests. Upon installation, it contacts `vendra[.]info` to retrieve a unique per-victim UUID and fetch a set of network rewriting rules that are refreshed hourly.

These rules are used to hijack and monetize search results in the victim's browser. When the victim performs an internet search, across any of more than 190 [Google country-code domains](#), the extension loads a script from `drelto[.]info/farlix` into the search results page. Because the script runs within the search engine's own origin context, it has full access to the page content. This enables the operator to inject advertisements into organic search results, rewrite affiliate links to capture referral revenue and/or redirect clicks to attacker-controlled destinations. These actions are functionally transparent to the user and visually indistinguishable from the legitimate search page.

The extension also checks `vendra[.]info` for updates via `/extensionInstaller/updateChromeExtension`, allowing the operator to silently rotate to a new extension version at any time. Install telemetry is reported to `finersto[.]com` and `extentrack[.]com`.

With more than 50 unique samples contacting `mqsearch[.]com` according to VirusTotal, Docro Hijacker appears to be a mature, independent monetization module.

Conclusion

The actors behind CL-CRI-1171 did not use sophisticated evasion techniques, but focused their efforts on building a loader that is exceptionally difficult to track due to its clever gating mechanisms.

Our investigation highlights how the authors of OfferLoader, by being intentional about its simplicity and by hiding all of its functions as bytecode within a package, designed the malware to evade scrutiny while quietly building a massive, mature infection funnel.

Although OfferLoader was easy for defenders and security products to miss, its gating mechanisms were not, and the sheer number of them was our first clue that more widespread activity was occurring.

While a loader that just drops a payload is seemingly routine, this mechanism allowed possibly thousands of rotational malware bundles, including entirely new and undocumented malware families, to remain hidden from standard security attention. Ultimately, this case serves as a critical reminder for defenders: Treating commodity loader infections as minor, routine events overlooks the dangerous payloads and campaigns that might be tied to them.

Palo Alto Networks Protection and Mitigation

Palo Alto Networks customers are better protected against the threats described in this report through the following products and services:

- The [Advanced WildFire](#) machine-learning models and analysis techniques have been reviewed and updated in light of the indicators shared in this research. Through continuous cloud-based analysis, Advanced WildFire is designed to proactively identify and block OfferLoader samples as well as downstream payloads, including Insomnia RAT, ARKTunnel, and Docro Hijacker.
- [Advanced URL Filtering](#) and [Advanced DNS Security](#) identify known domains and URLs associated with this activity as malicious.

- [Cortex XDR](#) and [XSIAM](#) can help detect and prevent the OfferLoader chain and all three payload branches described in this article. Cortex customers benefit from multiple layers of protection against this threat, including:
 - YARA-based signatures targeting the OfferLoader family and its staged payloads
 - Behavioral detection rules that help prevent:
 - Malicious Chrome extension setup
 - Untrusted service installations used for persistence
 - Trojanized installer execution patterns

If you think you may have been compromised or have an urgent matter, get in touch with the [Unit 42 Incident Response team](#) or call:

- North America: Toll Free: +1 (866) 486-4842 (866.4.UNIT42)
- UK: +44.20.3743.3660
- Europe and Middle East: +31.20.299.3130
- Asia: +65.6983.8730
- Japan: +81.50.1790.0200
- Australia: +61.2.4062.7950
- India: 000 800 050 45107
- South Korea: +82.080.467.8774

Palo Alto Networks has shared these findings with our fellow Cyber Threat Alliance (CTA) members. CTA members use this intelligence to rapidly deploy protections to their customers and to systematically disrupt malicious cyber actors. Learn more about the [Cyber Threat Alliance](#).

Indicators of Compromise

Initial Access and OfferLoader

SHA256 hash:

7f792c45de1e28fd42ac44c9444f157a2161742d130bac336c0e991aabb112c

File name: windirstat.exe

File type: PE32 executable, Inno Setup 6.7.1

File description: OfferLoader trojanized WinDirStat installer delivered via SEO-poisoning

SHA256 hash:

fc485882626512e7ff82a1d7cd8e8fb3e9751b026d97e682d6908aefff1f2d73

File name: windirstat.tmp

File type: PE32 executable, Inno Setup unpacked stage

File description: Unpacked WinDirStat stage

Operation A: Insomnia RAT

SHA256 hash:

3052bd320a34e12ee694811ed0578797477dfd480c664491e509ed15ce1a6961

File name: eld0.exe

File type: PE32 executable, Inno Setup 6.3.0 SetupLdr

File description: Insomnia RAT initial loader

SHA256 hash:

25558ea78c4aa0fdd0f45fafcaa546d3115dc5806809d144a5801a40e48fd4c5

File name: eld0.tmp

File type: PE32 executable, Inno Setup unpacked stage

File description: Unpacked loader stage

SHA256 hash:

9b0d9cbc0fd4a7bae8b78a15dfbe63052779414ad725845732c4a0083008da69

File name: a.dll

File type: PE32 DLL

File description: Executes the PowerShell downloader for the Node.js and Python second stages

SHA256 hash:

ceb30a5eb9ad9d9c6712c80726df16f96f99d9fc0753be241b00b4d636eb576e

File name: t.ps1

File type: PowerShell

File description: Insomnia RAT PowerShell loader

SHA256 hash:

cf184d04ca31fb2b6b7efd85399c29c1136b539153e137ceb3877b1b905791de

File name: <guid>.py

File type: Python script

File description: the Python-based component of the Insomnia RAT dual payload.

SHA256 hash:

62d49d0c78207ec2452cc8a30501db771c9edbae89889e41a7dd227551243e8e

File name: aa.js

File type: JavaScript

File description: the Node.js-based component of the Insomnia RAT dual payload.

URL: hxxps[:]//stryper[.]info/t.ps1

Description: Stage-2 PowerShell installer for the Node.js and Python agents

URL: hxxps[:]//stryper[.]info/aa.js

Description: Stage-3 Node.js agent

URL: hxxp[:]//aa.amazingshield[.]xyz/33244556546.py

Description: Stage-3 Python agent

Domain: stryper[.]info

Description: Second-stage PowerShell and Node.js agent host

Domain: aa.amazingshield[.]xyz

Description: Python agent host

Domain: crowdstri[.]com

Description: CrowdStrike-typosquat C2 for the Python agent

Operation B: ARKTunnel

SHA256 hash:

aaebc8c07de485be6d1bfa956668c5e18aa1ff5588dfe84672e20ae90b4560f1

File name: eld1.exe

File type: PE32 executable

File description: LSB-steganography dropper. The attacker forged the PE version-info resource to mimic a popular test proctoring application.

SHA256 hash:

e05bc22afbc5ddd50b49c85ee169dd13318000d38286de2b8bcff98217256a8d

File size: 92,657 bytes

File name: procorTrex.zip

File location: C:\Users\Public\procorTrex.zip

File type: ZIP archive

File description: ZIP carved from the eld1.exe BMP steganography payload

SHA256 hash:

b367762140ae7f5098230b8a5da738c9241f286281ec9439dd6ca581fc87989c

File size: 245,248 bytes

File name: wsc1.exe

File type: PE32 executable

File description: WebSocket tunneling RAT

ARKTunnel resource icons:

SHA256 hash:

d8d783f8e050a6e394f3c0aa5e2bc73a38d822e55fbc39c0648cbff566de3cdf

File description: Resource Icon shared across ARKTunnel samples

SHA256 hash:

06e0afd01bbc6c9d5dc16c3165089b233dc071e1f251abd06235d1b9166cdac5

File description: Resource Icon shared across ARKTunnel samples

Domain: reg.pcsdkflyer[.]ca

Description: wsc1.exe WebSocket RAT C2

Operation C: Docro Hijacker

SHA256 hash:

2c6e11027b011042c9a118fc20728f8f4ebb6be8795cc84cc9a45622c297d354

File name: eld2.exe

File type: PE32 executable, Inno Setup installer

File description: Branch C installer that drops Adblock.dll and the docro Chrome extension

SHA256 hash:

553ce594c9c6afdd4794fddc28c194e3bc3c1b052310e5099c58ac15bab72104

File name: eld2.tmp

File type: PE32 executable

File description: Inno Setup unpacked stage

SHA256 hash:

fdcc95b7791c0d6590dcf1a412dc9fcc92ad2095818d1b78368b31efad012007

File size: 3,041,280 bytes

File name: Adblock.dll

File location: %TEMP%\Adblock.dll

File type: PE32 DLL

File description: Chrome Secure Preferences HMAC-SHA256 bypass DLL that sideloads the docro extension and hijacks the default search provider

Docro extension path:

File location: C:\ProgramData\DocsHelper\docro

File description: Manifest V3 Chrome extension sideloaded by Adblock.dll

Domain: vendralo[.]info

Description: Per-victim UUID and hourly rule delivery

Domain: finersto[.]com

Description: Extension install and start telemetry

Domain: drelto[.]info

Description: SERP-injection script host

Domain: mqsearch[.]com

Description: Hijacked the default search provider endpoint

Domain: extentrack[.]com

Description: Docs Helper install-success and install-failure callback

CL-CRI-1171 Rotational Infrastructure

Initial-Access Lure and SEO File-Locker Hosts

Domain	Role
noiseship[.]cdfd	SEO-poison domain
atthelake[.]info	Top referring SEO-poison domain
uy.basesfiles[.]com	Fake file host
basesfile[.]com	Sibling fake file host

igk.filexspace[.]com ikx.filexspace[.]com	SEO file-locker lure hosts
filexstorage[.]site	SEO file-locker lure host
filesccloud[.]pro	SEO file-locker lure host
zippyfiles[.]net	SEO file-locker lure host
mifilesx[.]site	SEO file-locker lure host
dw.xrsdownload[.]com	SEO file-locker lure host
storage.ggclicker[.]com	SEO file-locker / referrer host
watchadvance[.]com	SEO-poisoning domain

Domains Used to Confirm OfferLoader Installations

Domain:

- animalview[.]xyz
- trickflag[.]info
- suitstraw[.]info

- connect.fuelleg[.]info
- vesselsystem[.]xyz
- minewave[.]info
- collartitle[.]info
- boardmagic[.]info
- placespoon[.]xyz
- needcherries[.]online

Payload-Handoff and Second-Stage Hosts

Install-Tracker Beacons (Operator Panel)

Domain	Role
voyagemist[.]space	PPI affiliate install tracker
statementtouch[.]xyz	PPI affiliate install tracker
chawton[.]info	PPI affiliate install tracker

YouTube Funnel – Burner Blogs and Custom-Domain Sites

Domain Name	Peers	Type
velfps.blogspot[.]com	V el v o x	Blogspot burner
velvoxlab.blogspot[.]com	V el v o x	Blogspot burner
venrx.blogspot[.]com	V e n r x	Blogspot burner

venrxhub.blogspot[.]com	V e n r x	Blogspot burner
venrx[.]xyz	V e n r x	Custom-domain burner
ravexoficial.blogspot[.]com	R a v e x	Blogspot burner
adex-blog.blogspot[.]com	A d e x	Blogspot burner

Please note: While Velvox is a real company, the attackers used its name coincidentally. They did not use or impersonate the company's resources or identity.

Additional Resources

- [“WinYahoo” PUP Modifies Chrome Secure Preferences](#) – Malwarebytes
- [NodeJS backdoors delivering proxyware and monetization schemes](#) – Walmart Global Tech blog

Appendix A: CL-CRI-1171 Full Discovered Infrastructure Tables

Indicator	Role
achievershelf[.]space	Gate/landing
activitymeal[.]space	Gate/landing
additionplot[.]cfd	Gate/landing
adviceturn[.]xyz	Gate/landing
afternoonscrew[.]space	Gate/landing
agreementjuice[.]space	Gate/landing
airplaneiron[.]xyz	Gate/landing

airtwig[.]xyz	Gate/landing
amountfuel[.]icu	Gate/landing
animalrecord[.]xyz	Gate/landing
apparatusruck[.]xyz	Gate/landing
apparelplate[.]space	Gate/landing
archairport[.]xyz	Gate/landing
authoritykittens[.]info	Gate/landing
babyvein[.]xyz	Gate/landing
badgewing[.]xyz	Gate/landing
bagcare[.]space	Gate/landing
basinpleasure[.]xyz	Gate/landing

basketball year[.]xyz	Gate/landing
baskethumo r[.]xyz	Gate/landing
bedroomdes ire[.]xyz	Gate/landing
beliefpict ure[.]xyz	Gate/landing
bellplaygr ound[.]xyz	Gate/landing
bikesdonke y[.]info	Gate/landing
birthdayma gic[.]xyz	Gate/landing
boatthough t[.]xyz	Gate/landing
boundarych ickens[.]x yz	Gate/landing
boytank[.] xyz	Gate/landing
branchmorn ing[.]xyz	Gate/landing

breathdoctor[.]xyz	Gate/landing
bubbleslip[.]xyz	Gate/landing
cabbagemeasure[.]xyz	Gate/landing
cablename[.]xyz	Gate/landing
cardgrape[.]xyz	Gate/landing
cattlegold[.]xyz	Gate/landing
celeryerror[.]xyz	Gate/landing
centscarf[.]xyz	Gate/landing
chalkprose[.]xyz	Gate/landing
cherriestruck[.]space	Gate/landing
chesstail[.]xyz	Gate/landing

chickensmine[.]space	Gate/landing
churchpail[.]xyz	Gate/landing
clothcrib[.]xyz	Gate/landing
clothcurrent[.]xyz	Gate/landing
coatberry[.]xyz	Gate/landing
connect.activitykitty[.]xyz	Install tracker
connect.aparatustaste[.]xyz	Install tracker
connect.armcard[.]xyz	Install tracker
connect.badgeterritory[.]xyz	Install tracker

connect.ba itmetal[.] xyz	Install tracker
connect.be efteeth[.] xyz	Install tracker
connect.be lievesiste rs[.]xyz	Install tracker
connect.bo undaryfly[.]xyz	Install tracker
connect.bu bbleapplia nce[.]xyz	Install tracker
connect.ca bleland[.] xyz	Install tracker
connect.ch inexpert[.]xyz	Install tracker
connect.co nditiongra de[.]xyz	Install tracker

connect.co ppersummer [.]xyz	Install tracker
connect.cr eatorcreat or[.]xyz	Install tracker
connect.dr esstent[.] xyz	Install tracker
connect.dr opjeans[.] xyz	Install tracker
connect.ed geplaygrou nd[.]xyz	Install tracker
connect.ex changeclub [.]xyz	Install tracker
connect.ex istencedis cussion[.] info	Install tracker
connect.ex pansional t[.]info	Install tracker

<code>connect.fangstitch[.]]xyz</code>	Install tracker
<code>connect.fogparcel[.] info</code>	Install tracker
<code>connect.fodhook[.]i nfo</code>	Install tracker
<code>connect.forkcountry[.]xyz</code>	Install tracker
<code>connect.ge esepurpose [.]xyz</code>	Install tracker
<code>connect.gi antsdogs[.] info</code>	Install tracker
<code>connect.gi raffetooth paste[.]xy z</code>	Install tracker
<code>connect.gu itarrobin[.]xyz</code>	Install tracker

connect.ha lllunch[.] info	Install tracker
connect.ha rborclam[.] xyz	Install tracker
connect.ho lecompany[.]info	Install tracker
connect.kn ifesea[.]i cu	Install tracker
connect.mo nthsmoke[.]info	Install tracker
connect.no segovornor [.]xyz	Install tracker
connect.pa gesubstanc e[.]xyz	Install tracker
connect.po isonblade[.]xyz	Install tracker

connect.pr osesalt[.] xyz	Install tracker
connect.pu rposething s[.]info	Install tracker
connect.qu iltgirls[.] xyz	Install tracker
connect.re actionbit[.]info	Install tracker
connect.re wardrun[.] xyz	Install tracker
connect.ri ceapparel[.]xyz	Install tracker
connect.se ashorelett ers[.]info	Install tracker
connect.se lectiondog s[.]xyz	Install tracker

connect.sh apeboot[.] info	Install tracker
connect.sh optax[.]xy z	Install tracker
connect.st opfinger[.] info	Install tracker
connect.st ructurekis s[.]xyz	Install tracker
connect.ta nksuggesti on[.]xyz	Install tracker
connect.th ingbrass[.] xyz	Install tracker
connect.th oughtslave [.]xyz	Install tracker
connect.th readfuel[.] xyz	Install tracker

connect.tr ailcontrol [.]xyz	Install tracker
connect.tr ickbushes[.]info	Install tracker
connect.va cationthou ght[.]xyz	Install tracker
connect.vi ewschoo[.]xyz	Install tracker
connect.vo yage[.]xyz	Install tracker
connect.wo olreward[.]xyz	Install tracker
connect.ye aricicle[.]xyz	Install tracker
connect.ze bratranspo rt[.]xyz	Install tracker

controlprice[.]xyz	Gate/landing
coughcoal[.]icu	Gate/landing
countrypipe[.]space	Gate/landing
cowsfoot[.]xyz	Gate/landing
crackfood[.]space	Gate/landing
creamfurniture[.]space	Gate/landing
creditchickens[.]xyz	Gate/landing
creditededucation[.]cfd	Gate/landing
crimestreet[.]xyz	Gate/landing
crimesupport[.]cfd	Gate/landing
curvebite[.]xyz	Gate/landing

deathrock[.]xyz	Gate/landing
deathshop[.]xyz	Gate/landing
decisionre action[.]x yz	Gate/landing
dinosaursj am[.]cfd	Gate/landing
distancebe droom[.]xy z	Gate/landing
distributi ontheory[.]cfd	Gate/landing
dolldebt[.]xyz	Gate/landing
doorsoap[.]cfd	Gate/landing
dustprotes t[.]icu	Gate/landing
expansions alt[.]info	Gate/landing

fallbeginner[.]xyz	Gate/landing
fangbear[.]xyz	Gate/landing
filescenter[.]cloud	Gate/landing
filesilo[.]cloud	Gate/landing
fingerbasketball[.]xyz	Gate/landing
flavorwood[.]xyz	Gate/landing
fleshfrog[.]xyz	Gate/landing
fleshplants[.]xyz	Gate/landing
fleshproduce[.]xyz	Gate/landing
foodrock[.]space	Gate/landing
forkmice[.]xyz	Gate/landing

friendjewel[.]cfd	Gate/landing
geeseairport[.]xyz	Gate/landing
girlsgrain[.]xyz	Gate/landing
glassmove[.]xyz	Gate/landing
goldsteel[.]cfd	Gate/landing
governmentyard[.]cfd	Gate/landing
grandfatherquiver[.]xyz	Gate/landing
gripcollar[.]xyz	Gate/landing
gripsleep[.]xyz	Gate/landing
gunbear[.]xyz	Gate/landing
hairreward[.]xyz	Gate/landing

hatescale[.]info	Gate/landing
healthiron [.]space	Gate/landing
holemuscle [.]xyz	Gate/landing
homecub[.]cfd	Gate/landing
homefireman[.]xyz	Gate/landing
honeyfear[.]xyz	Gate/landing
ilesilo[.]cloud	Gate/landing
instrument volcano[.]space	Gate/landing
kittenschal[.]xyz	Gate/landing
kittensgrade[.]cfd	Gate/landing
kittensrobin[.]info	Gate/landing

landerror[.]xyz	Gate/landing
lesilo[.]cloud	Gate/landing
liptendency[.]info	Gate/landing
liquidtoes[.]xyz	Gate/landing
liquidwrench[.]cfd	Gate/landing
lockettrail[.]xyz	Gate/landing
lumberbaseball[.]xyz	Gate/landing
memorycompany[.]xyz	Gate/landing
micesisters[.]xyz	Gate/landing
milkname[.]xyz	Gate/landing
minuteblade[.]xyz	Gate/landing

mountainsu rprise[.]c fd	Gate/landing
mouthfruit [.]cfd	Gate/landing
noiseship[.]cfd	Gate/landing
northbox[.]xyz	Gate/landing
partpipe[.]xyz	Gate/landing
partyfrien ds[.]cfd	Gate/landing
passengerb rake[.]spa ce	Gate/landing
peacejewel [.]xyz	Gate/landing
peacetongu e[.]xyz	Gate/landing
petministe r[.]xyz	Gate/landing
picturepor ter[.]cfd	Gate/landing

pieplant[.]space	Gate/landing
pizzasthread[.]xyz	Gate/landing
pleasurewaves[.]info	Gate/landing
popcornregret[.]xyz	Gate/landing
porterdebt[.]xyz	Gate/landing
powerbuses[.]xyz	Gate/landing
profitfact[.]xyz	Gate/landing
prosetoothbrush[.]xyz	Gate/landing
quincepage[.]xyz	Gate/landing
raintexture[.]xyz	Gate/landing
rakepurpose[.]xyz	Gate/landing

raterake[.]cfd	Gate/landing
readingsci ence[.]xyz	Gate/landing
reasonachi ever[.]xyz	Gate/landing
recessgira ffe[.]xyz	Gate/landing
recordhist ory[.]xyz	Gate/landing
regretsqui rrel[.]xyz	Gate/landing
restbucket [.]xyz	Gate/landing
ricestar[.]xyz	Gate/landing
rifledog[.]xyz	Gate/landing
ringsparce l[.]xyz	Gate/landing
roadyear[.]xyz	Gate/landing
rockcredit [.]space	Gate/landing

rollglass[.]xyz	Gate/landing
roofbattle[.]xyz	Gate/landing
roofreaction[.]xyz	Gate/landing
rosegrip[.]xyz	Gate/landing
routeletters[.]xyz	Gate/landing
runhouses[.]xyz	Gate/landing
scarecrowcare[.]xyz	Gate/landing
scentrod[.]info	Gate/landing
screwbirth[.]xyz	Gate/landing
seatlace[.]space	Gate/landing
seaword[.]xyz	Gate/landing
servantadvice[.]xyz	Gate/landing

shiptank[.]cfd	Gate/landing
shirtexamp le[.]xyz	Gate/landing
shoesearth quake[.]xy z	Gate/landing
sinkwash[.]space	Gate/landing
skirtloss[. .]xyz	Gate/landing
slipvegeta ble[.]xyz	Gate/landing
smokecar[.]space	Gate/landing
songtheory [.]xyz	Gate/landing
spadeleg[.]xyz	Gate/landing
sparkrice[. .]space	Gate/landing
sparkrub[.]xyz	Gate/landing

spoonducks [.]cfd	Gate/landing
springdogs [.]xyz	Gate/landing
spybasebal l[.]space	Gate/landing
startmonke y[.]cfd	Gate/landing
statements ervant[.]x yz	Gate/landing
statementt ouch[.]xyz	Gate/landing
steamhouse s[.]cfd	Gate/landing
stopzinc[.]]xyz	Gate/landing
structurel inen[.]xyz	Gate/landing
suggestion cemetery[.]]xyz	Gate/landing
suitsoap[.]]xyz	Gate/landing

suitstraw[.]info	Gate/landing
swimrest[.]xyz	Gate/landing
tablechess [.]info	Gate/landing
tailsilk[.]xyz	Gate/landing
territoryc aption[.]x yz	Gate/landing
texturebad ge[.]xyz	Gate/landing
thrillduck s[.]xyz	Gate/landing
throneback [.]xyz	Gate/landing
tinsofa[.] xyz	Gate/landing
toespiders [.]xyz	Gate/landing
toothpaste sense[.]xy z	Gate/landing

toothpaste sun[.]xyz	Gate/landing
townquiver [.]xyz	Gate/landing
trampdonke y[.]icu	Gate/landing
treesboard [.]xyz	Gate/landing
truckpig[.]]cfd	Gate/landing
truckshat[.]xyz	Gate/landing
turnclass[.]xyz	Gate/landing
umbrellave ssel[.]xyz	Gate/landing
vacationen gine[.]xyz	Gate/landing
vesselsyst em[.]xyz	Gate/landing
vestthings [.]cfd	Gate/landing
volcanopin [.]xyz	Gate/landing

voyagemist [.]space	Gate/landing
wastereadi ng[.]xyz	Gate/landing
wasteturke y[.]xyz	Gate/landing
wastewine[.]xyz	Gate/landing
wavepan[.] xyz	Gate/landing
whistleboo k[.]cfd	Gate/landing
whistleson g[.]xyz	Gate/landing
wormspark[.]xyz	Gate/landing
woundsecre tary[.]xyz	Gate/landing
wrenobserv ation[.]xy z	Gate/landing
yamsmell[.]xyz	Gate/landing

yardvalue[.]cfd	Gate/landing
zephyrhall[.]cfd	Gate/landing

Appendix B: YouTube Channels Linked to CL-CRI-1171 Activity

(Channels were taken down after we notified Google.)

Pers ona	Channel
Velv ox	@VelvoxYT
Venr x	@Venrx
Rave x	@RAVEX-wu1pg
Ade x	@adex915
Ripe x	@ripex

HAS NZ TW EAK S	@hasnzyt
Ontr endy tt	@OnTrendd
Tren d Rise	@trendrise
Rekn otic Lab	@Reknotic
Now FixT utori als	@NowFixTutorials
f4fix	@f4fix